Compositional Reasoning Depth Predicts Clinical AI Failure: Empirical Evidence Consistent with the Peng Impossibility Theorem in Electronic Health Record Question Answering

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Abstract

Aggregate accuracy benchmarks conceal a systematic structure in clinical AI failure: questions requiring more inferential steps produce disproportionately more errors. Motivated by the Peng impossibility theorem for transformer attention composition, we introduce a pre-registered *hop-count taxonomy*—the number of distinct reasoning steps required to answer a clinical question from an electronic health record (EHR)—as a principled predictor of model failure. We annotate 313 clinician-generated EHR question-answer pairs from MedAlign across four hop levels and evaluate 301 questions in a within-model ablation (`claude-sonnet-4-6`, zero-shot versus extended thinking) and cross-architecture replications (`gpt-4o` and `gpt-5.4-2026-03-05`, zero-shot). All three model families exhibit monotone accuracy decline with hop count: Claude Sonnet zero-shot declines from 30.6% (hop=1) to 17.6% (hop=4) (Cochran–Armitage $z = -2.30$, $p = 0.011$; OR per hop $= 0.72$, 95% CI [0.56, 0.92], $p = 0.008$); GPT-4o independently replicates this pattern (37.8% to 14.7%; CA $z = -3.80$, $p < 0.0001$; OR $= 0.58$ [0.45, 0.75], $p < 0.001$); GPT-5.4 confirms the pattern in the current flagship model (37.8% to 23.5%; CA $z = -2.22$, $p = 0.013$; OR $= 0.80$ [0.66, 0.98], $p = 0.027$). A pre-specified context sufficiency audit confirmed that higher-hop questions are not differentially disadvantaged by EHR truncation (answerability 93–95% at hops 2–4 vs. 79% at hop=1), establishing that the degradation reflects compositional reasoning difficulty. Extended thinking did not significantly flatten the accuracy-versus-depth curve across three reasoning conditions (internal extended thinking, structured four-step CoT, and 5× larger thinking budget), and thinking-token usage scaled with hop count ($r = 0.31$, $p < 0.0001$), consistent with the predicted $O(k)$ computational requirement. These findings establish hop count as a theoretically grounded, cross-architecture predictor of clinical AI failure with immediate implications for deployment risk stratification.

1 Introduction

Artificial intelligence systems based on transformer architectures are now embedded in clinical workflows ranging from automated chart summarization to real-time diagnostic support [Singhal et al., 2023, McDuff et al., 2025]. Despite rapidly improving benchmark performance on standardized medical examinations [Jin et al., 2021, Hendrycks et al., 2021], these systems exhibit inconsistent and

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sometimes dangerous failures when applied to real-world electronic health record (EHR) tasks [Wu et al., 2025]. A persistent challenge is that aggregate accuracy metrics conceal the structure of failure: researchers and clinicians typically observe *that* a model fails without understanding *why* or *when* it will fail.

We propose that compositional reasoning depth—the number of distinct inferential hops required to produce a correct answer—is a principled predictor of clinical AI failure, grounded in a formal theoretical result. Peng et al. [2024] prove an impossibility theorem for transformer self-attention: a single attention layer with H heads, head dimension d, and numerical precision p cannot solve the (n, k)-function-composition problem when $n \log n > H \cdot d \cdot p$. Informally, once the context length n exceeds a threshold determined by the model's architectural parameters, the attention mechanism lacks sufficient representational capacity to chain multiple inferential steps within a single forward pass. Three theoretical escape routes exist: multi-step chain-of-thought reasoning (which effectively adds layers), retrieval-augmented generation, and non-standard architectural choices such as state-space models or mixture-of-expert routing [Zhou et al., 2023, Yehudai et al., 2025]. While the theorem formally characterizes single-layer attention, practical transformers stack many such layers; we therefore treat it as a theoretical framework motivating *qualitative* predictions about compositional difficulty rather than applying it quantitatively to the multi-layer setting. The critical empirical question is whether accuracy degrades with hop count and whether test-time reasoning provides relief, predictions that hold regardless of whether the single-layer impossibility is the proximate mechanism.

Clinical EHR question answering is a natural testing ground for this theorem: retrieving a single fact (hop=1), computing a temporal overlap (hop=3), or synthesizing heterogeneous entries across domains (hop=4) constitute structurally distinct tasks predicted to exhibit qualitatively different error rates, with test-time reasoning predicted to selectively benefit higher-hop tasks.

This paper makes six contributions: (1) a pre-registered hop-count taxonomy for clinical EHR QA annotating 313 MedAlign questions across four hop levels; (2) cross-architecture empirical evidence of hop-count-predicted failure in EHR QA across three model families (`claude-sonnet-4-6`, `gpt-4o`, and `gpt-5.4-2026-03-05`); (3) a context sufficiency audit ruling out EHR truncation as a confound; (4) mechanistic evidence of $O(k)$ computation scaling ($r = 0.31$, $p < 0.0001$); (5) characterization of error structure by hop level showing that structured decomposition converts hallucination to omission; and (6) question-type sensitivity analysis and truncation robustness ablation confirming the hop–accuracy relationship is not confounded (adjusted OR = 0.57, $p = 0.005$).

2 Background and Related Work

The Peng impossibility theorem. Peng et al. [2024] establish that for any single-layer transformer attention mechanism, the (n, k)-function-composition problem—which asks whether the model can compose k arbitrary functions over a context of length n—is unsolvable when $n \log n > H \cdot d \cdot p$. This is not an empirical observation but a formal impossibility result derived from communication complexity theory. The theorem applies to the softmax self-attention operation as standardized in modern transformers and does not depend on training procedure, model scale, or the specific clinical domain. Critically, the theorem identifies test-time chain-of-thought reasoning as a theoretically sound mitigation: by decomposing a k-hop problem into k sequential single-hop steps, each within the computability threshold, CoT reasoning should flatten the accuracy degradation curve. Yehudai et al. [2025] provide complementary analysis showing that recurrent architectures and chain-of-thought can solve compositional tasks that standard transformers cannot, independently corroborating the Peng escape route. Zhou et al. [2023] demonstrate architectural alternatives (non-uniform block structures) that can route around the attention bottleneck, though at engineering complexity cost.

Multi-hop reasoning benchmarks. The NLP community has extensively studied multi-hop reasoning in open-domain QA settings. HotpotQA [Yang et al., 2018] provides 113K questions requiring evidence synthesis from two Wikipedia paragraphs; MuSiQue [Trivedi et al., 2022] constructs up to 4-hop questions by composing single-hop questions with explicit bridge entities; 2WikiMultihopQA [Ho et al., 2020] adds cross-document coreference. Across these benchmarks, accuracy declines consistently with the number of required hops, attributed to error propagation and attention capacity limitations. However, no prior study has (a) grounded the hop–accuracy relationship in a formal theorem, (b) tested the theoretical CoT mitigation as a function of hop depth rather than on average, or (c) applied hop-count analysis to clinical EHR data.

Clinical AI benchmarks and EHR reasoning. MedQA [Jin et al., 2021] and MMLU Medical [Hendrycks et al., 2021] evaluate LLMs on medical licensing examinations—closed-book multiple-choice requiring medical knowledge but not EHR navigation. Med-PaLM [Singhal et al., 2023] achieved expert-level performance on such benchmarks but was not evaluated on compositional EHR reasoning. emrQA [Pampari et al., 2018] provides a large EHR QA benchmark but questions are primarily single-hop slot-filling with no compositional depth taxonomy. Clinical TempEval [Bethard et al., 2016] establishes temporal relation extraction as a distinct challenging subtask in clinical NLP, consistent with the temporal arithmetic component of our hop=3 category. FDARxBench [Xiong et al., 2026] shows retrieval and long-context grounding remain bottlenecks in document-grounded biomedical QA—consistent with our finding that the hop–accuracy relationship is not resolved by more context. MedAlign [Fleming et al., 2024] provides clinician-generated, open-ended questions paired with EHR XML records and clinician reference answers, making it the most ecologically valid benchmark for testing compositional EHR reasoning. No prior study has stratified MedAlign performance by compositional depth.

Chain-of-thought and reasoning models. Wei et al. [2022] demonstrated that chain-of-thought prompting elicits multi-step reasoning in large models; Kojima et al. [2022] showed a zero-shot CoT trigger is sufficient. Nye et al. [2021] provided theoretical grounding through scratchpads as intermediate computation. Reasoning-specialized models—including OpenAI o1 [OpenAI, 2024] and DeepSeek R1 [DeepSeek-AI, 2025]—extend these ideas through reinforcement learning over extended thinking traces. Merrill and Sabharwal [2024] prove that transformers with polynomially many CoT tokens can solve problems inaccessible to direct inference, with required token count scaling with reasoning depth—a formal analog of the $O(k)$ computation scaling our P3 test examines empirically. No study has measured CoT benefit as a function of hop depth in clinical settings.

Safety and variance in clinical AI. Wu et al. [2025] document query-type-dependent hallucination rates in clinical LLMs without compositional depth stratification. Prior work on automation bias [Goddard et al., 2012, Lyell and Coiera, 2017] has not examined hop-count moderation. The error type structure—hallucination versus omission versus reasoning error—as a function of compositional depth has not been characterized.

3 Problem Formulation

The Peng theorem and clinical QA. Let $\mathcal{E}$ denote the EHR context of length n tokens, and let $\mathcal{Q}$ denote a clinical question requiring composition of k functions $f_1, \ldots, f_k$ over $\mathcal{E}$, where each f_i retrieves or transforms a specific subset of the EHR. The (n, k)-function-composition problem asks a model to produce a correct answer by sequentially applying these functions. By the Peng theorem [Peng et al., 2024], a single attention layer with parameters (H, d, p) cannot solve this problem when $n \log n > H \cdot d \cdot p$. For contemporary large-context models serving long EHRs, n is on the order of 10^3–10^4 tokens, making the left-hand side $n \log n \approx 10^4$–10^5. Typical model architectural parameters place $H \cdot d \cdot p$ in a range that, while large, is finite and potentially crossed by sufficiently complex EHR reasoning tasks.

The theorem generates three testable predictions: (P1) *Monotone degradation*: Accuracy should decrease monotonically with k under standard (non-CoT) inference. (P2) *CoT flattening*: Chain-of-thought reasoning, by decomposing k-hop problems into k sequential 1-hop steps, should flatten the accuracy-versus-hop curve, with greater absolute benefit at higher k. (P3) *Computation scaling*: The amount of reasoning computation (proxied by thinking token count in extended-thinking models) should scale with k, consistent with the theoretical requirement for $O(k)$ sequential computations.

Scope of the theoretical mapping. The Peng theorem characterizes a *single* attention layer; modern transformers stack many such layers, which can in principle decompose reasoning through depth. We interpret P1–P3 as qualitative predictions motivated by the theorem rather than strict corollaries. The empirical claims stand independently: compositional depth predicts accuracy, thinking tokens scale with depth, and test-time reasoning does not flatten the degradation curve at current budgets.

Hop-count taxonomy. We define a pre-registered four-level hop taxonomy (Table 1): hop=1 (single-fact retrieval), hop=2 (two-fact composition or threshold comparison), hop=3 (three-step

Table 1: Pre-registered hop-count taxonomy for clinical EHR question answering, with operational definitions and representative examples from MedAlign.

Hop	Operational Definition	Representative Example
1	Direct retrieval of a single named fact from the EHR.	"What is the patient's current allergy list?"
2	Two-fact composition, threshold application, or binary temporal comparison.	"Did the patient have an elevated WBC on admission?"
3	Three-step reasoning chain; requires cross-domain lookup, temporal arithmetic, or contextual clinical guideline application.	"Is this patient eligible for supplemental breast cancer screening based on current guidelines?"
4	Multi-component synthesis or generation requiring integration of $\geq$4 EHR sources or time spans.	"Summarize all specialist consultations and flag unresolved recommendations."

inference chain: cross-domain retrieval, temporal reasoning, or guideline application), and hop=4 (multi-component synthesis or generation across $\geq$4 EHR sources or time spans).

4 Methods

Dataset: MedAlign. MedAlign [Fleming et al., 2024] consists of 983 clinician-generated instructions paired with EHR XML files from the Stanford STARR-OMOP database and clinician reference answers. We use the publicly released subset of 402 unique question-answer pairs for which clinician-reviewed model responses and source EHR files are available. After restricting to questions with a uniquely identifiable EHR file and valid clinician reference response, 313 questions were annotated and eligible; of these, 301 had complete model evaluations (API errors or timeouts accounted for 12 omissions), yielding the final analytic sample (hop=1: $n = 111$, hop=2: $n = 46$, hop=3: $n = 42$, hop=4: $n = 102$; 602 total evaluations across both conditions). EHR records were parsed from XML, chronologically structured, and truncated to 8,000 characters (approximately 1,600–2,000 tokens at 4–5 characters per token, well within the model's 200K-token context window) while retaining the full character count as a completeness covariate.

Context sufficiency analysis. To test whether EHR truncation differentially disadvantaged higher-hop questions, we computed evidential coverage for each question; full methodology and thresholds are in Appendix F.

Hop annotation. All 313 questions were annotated with hop count prior to any model evaluation (pre-registration; annotation log and timestamp available in supplementary materials). Each question was presented to an LLM annotator together with a structured rubric defining each hop level, with explicit examples and a chain-of-thought elicitation prompt. To assess annotation reliability, a second independent annotation pass was run on a stratified random sample of 50 questions (12–13 per hop level) using the same model and prompt (within-model reproducibility), and a cross-model validation was performed using `gpt-4o`. Within-model reproducibility was substantial ($\kappa = 0.787$, 95% CI [0.603, 0.970]; 84% exact agreement, 98% agreement within $\pm$1 hop). Cross-model agreement was moderate ($\kappa = 0.464$, 95% CI [0.292, 0.637]; 60% exact). All annotations were finalized and locked before any model inference was run, precluding post-hoc adjustment.

Three board-certified clinicians (internist, radiologist, cardiologist) independently annotated the same 50-question sample to validate the LLM labels. Majority-vote agreement with Claude was $\kappa = 0.916$ (95% CI [0.71, 1.00]; 47/50, 94% exact), confirming near-perfect convergence with expert judgment; mean pairwise clinician–clinician $\kappa = 0.487$ is consistent with multi-hop benchmark standards, with disagreements at the hop=2/3 boundary. Clinician majority–GPT-4o $\kappa = 0.311$ confirms the moderate cross-model $\kappa = 0.464$ reflects GPT-4o calibration divergence, not taxonomy instability. Full statistics and CA trend replication under clinician labels are in Appendix B.

Experimental design. We evaluate two conditions using `claude-sonnet-4-6`, a publicly available large language model with extended-thinking capability:

- **Condition A (Zero-Shot):** The model receives a patient EHR and question with the prompt "Answer concisely" and no explicit reasoning instructions. This is the dense baseline.
- **Condition B (Extended Thinking):** The same model receives the same prompt with extended thinking enabled (`thinking={"type":"enabled","budget_tokens":3000}`), allowing an internal chain-of-thought scratchpad before producing the visible response.

This within-model ablation design isolates the effect of test-time reasoning from confounds such as model capacity, training data, or architectural differences. Both conditions use identical weights, identical prompts, and identical context. The sole manipulation is whether reasoning is enabled. We pre-registered this design and judge prompt (see Section 4) before observing any experimental results.

Automated judge. Responses were evaluated by a pre-registered automated judge using `gpt-4o-mini` with the following system prompt: *"You are a clinical QA evaluator. Score model responses against clinician reference answers. Return ONLY valid JSON: {'correct': true or false, 'error_type': 'none' | 'omission' | 'hallucination' | 'reasoning_error', 'confidence': 0.0–1.0}. Correct means the model response substantially agrees with the clinician reference. Omission means the model failed to answer, refused, or gave an empty or irrelevant response. Hallucination means the model stated facts not present in the EHR or contradicting established medical knowledge. Reasoning_error means the model had the right facts but reached the wrong conclusion."* This binary scoring schema was adopted after an initial pilot with a 4-category schema revealed systematic under-scoring (models rated "partial" rather than "correct" were all coded as incorrect), collapsing true accuracy. Physician adjudication of 100 evaluations (16% sample) stratified by hop level and judge confidence was used to validate judge calibration. The sample size was determined by power analysis: to achieve a 95% confidence interval for Cohen's κ that excludes the "acceptable" threshold of 0.60 when true agreement is $\kappa = 0.80$, a minimum of 100 cases is required (estimated CI width $\approx \pm 0.13$ at $n = 100$ [Donner and Eliasziw, 1992]). Stratification allocates 25 cases per hop level, with oversampling of cases where judge confidence < 0.70 (the borderline stratum) within each hop stratum; both experimental conditions are represented. Physician adjudication was completed by two board-certified clinicians (P1, lenient criterion; P2, strict criterion) and validated the AI judge against the strict standard: $\kappa(\text{AI}, \text{P2}) = 0.957$ (95% CI [0.874, 1.000]), exceeding the pre-registered threshold of $\kappa \geq 0.70$ (see Appendix C).

Statistical analysis. The pre-registered primary analysis specified a GLMM; MAP-based estimation (`BinomialBayesMixedGLM`) showed optimization instability with 301 parameters observed only twice each, so the pre-specified GEE fallback (logistic, independence working correlation) is the primary reported model. Predictors: hop count, condition, their interaction, EHR character count, and question token count; the $\beta_{\text{interaction}}$ term is the primary P2 test. We additionally report Cochran–Armitage trend tests (one-tailed, $\alpha = 0.05$) and an individual-level Pearson r for P3 (hop vs. thinking tokens). All analyses were finalized prior to model evaluation.

Question-type sensitivity analysis. To address question-type confounding, we extended the GEE model with reasoning-type dummies (factual/comparative/generative/computational; factual as reference) and ran a within-category Cochran–Armitage test for the comparative reasoning type (hops 2–4, $n = 16, 30, 20$; Appendix D).

Thinking-token computation. Extended-thinking token counts were obtained from the provider's API response as the sum of the `thinking` block lengths in each response, approximated as (word count) $\times 1.3$ to convert from approximate words to tokens. Raw thinking blocks are not retainable under the provider's terms of service; we release the token-count column in the public data release to enable cross-backend normalization by future investigators.

Truncation robustness ablation. To test whether P1 persists with larger EHR context, we run an additional zero-shot condition (*ZS-32K*) with an 8,000-character truncation limit replaced by 32,000 characters (a $4\times$ expansion), covering an estimated 6,400–8,000 tokens. All 301 questions are re-evaluated under this condition with identical prompting, judge, and scoring. Results are reported in Section 5 alongside the primary analyses.

Table 2: Error type distribution at hop=3 for zero-shot and CoT conditions in the pre-registered validation run ($n = 22$ questions per condition). CoT reduces hallucination but increases omission, consistent with the model learning to recognize its own uncertainty.

Condition	**Correct**	**Hallucination**	**Omission**	**Reasoning Error**	**Accuracy**
Zero-shot	5	13	3	1	22.7%
CoT	7	10	5	0	31.8%
Δ (CoT − ZS)	+2	−3	+2	−1	**+9.1 pp**

Table 3: Accuracy by condition and hop count ($n = 301$ questions). Cells show accuracy (%) with 95% Wilson confidence intervals. The GPT-4o replication used identical questions, EHR context, and judge. GEE per-hop OR quantifies accuracy degradation per additional reasoning hop in each condition.

	Hop Count						
Condition	**1** (n=111)	**2** (n=46)	**3** (n=42)	**4** (n=102)	**Overall**	**OR/hop**	p
Claude ZS	30.6 [22.8, 39.7]	28.3 [17.3, 42.5]	21.4 [11.7, 35.9]	17.6 [11.5, 26.2]	24.6 [20.1, 29.7]	0.72 [0.56, 0.92]	0.008
Claude ET	34.2 [26.1, 43.5]	30.4 [19.1, 44.8]	19.0 [10.0, 33.3]	19.6 [13.1, 28.4]	26.6 [21.9, 31.8]	0.70 [0.55, 0.88]	0.003
GPT-4o ZS*	37.8 [29.4, 47.1]	26.1 [15.6, 40.3]	23.8 [13.5, 38.5]	14.7 [9.1, 22.9]	26.2 [21.5, 31.5]	0.58 [0.45, 0.75]	<0.001
GPT-5.4 ZS*	37.8 [29.4, 47.1]	28.3 [17.3, 42.5]	28.6 [17.2, 43.6]	23.5 [16.4, 32.6]	30.2 [25.3, 35.6]	0.80 [0.66, 0.98]	0.027
Δ GPT-4o − Claude ZS	+7.2 pp	−2.2 pp	+2.4 pp	−2.9 pp	+1.6 pp	—	—

Accuracy (%) with 95% Wilson CI in brackets. OR/hop from GEE logistic regression with independence working correlation (full interaction model; ET slope computed via delta method as $\hat{\beta}_{\text{hop}} + \hat{\beta}_{\text{interaction}}$). *Independent replication: identical questions, EHR context, and judge; `gpt-5.4-2026-03-05` (pinned).

5 Results

Preliminary validation (hop=3). A pre-registered validation run on 22 hop=3 questions found zero-shot accuracy 22.7% and CoT accuracy 31.8% (+9.1 pp; Table 2). CoT reduced hallucinations (13→10) but slightly increased omissions (3→5), consistent with the mechanism proposed by Nye et al. [2021]: explicit reasoning causes the model to recognize EHR insufficiency rather than confabulate. This hallucination-to-omission shift has favorable clinical safety implications.

Full experiment results. Across 602 evaluations, zero-shot accuracy declined monotonically from 30.6% (hop=1) to 17.6% (hop=4; Table 3, Figure 1). Extended thinking followed a similar slope (34.2%, 30.4%, 19.0%, 19.6%), with a marginal 2.1 pp overall advantage; at hop=3 extended thinking slightly underperformed zero-shot (19.0% vs. 21.4%).

Primary outcome: hop × condition interaction. Cochran–Armitage trend tests confirmed monotone accuracy decline for both conditions (ZS: $z = -2.30$, $p = 0.011$; ET: $z = -2.62$, $p = 0.004$). GEE logistic regression showed each additional hop associated with a 28% odds reduction under ZS (OR = 0.72, 95% CI [0.56, 0.92], $p = 0.008$) and 30% under ET (OR = 0.70 [0.55, 0.88], $p = 0.003$), confirming P1. The main effect of ET did not reach significance (OR = 1.18, 95% CI [0.85, 1.65], $p = 0.32$). The hop × condition interaction (P2 test) was not significant (OR = 0.97 [0.84, 1.13], $p = 0.72$), and EHR length covariates were non-significant ($p = 0.67$, $p = 0.24$). These estimates were robust to working-correlation specification: an exchangeable-structure GEE yielded identical coefficients (hop OR = 0.72 [0.56, 0.92], $p = 0.008$; interaction OR = 0.97 [0.84, 1.13], $p = 0.72$).

Question-type sensitivity analysis. Adjusting for reasoning type strengthened the hop OR (0.72 unadjusted → 0.57, 95% CI [0.38, 0.84], $p = 0.005$), indicating question-type composition does not explain the hop effect. Within-comparative-type CA trend ($z = -1.90$, $p = 0.029$) confirmed P1 within a single reasoning category (hops 2–4, $n = 16$, 30, 20; Appendix D).

Computation scaling (P3). Mean thinking-token usage increased monotonically: 69.8 (hop=1), 83.0 (hop=2), 127.0 (hop=3), 135.7 (hop=4); individual-level Pearson $r = 0.31$ ($p < 0.0001$, $n = 301$, Figure 2), confirming P3. Mean usage (70–136 tokens) was far below the 3,000-token ceiling, ruling out budget exhaustion as an explanation for the P2 null result. A pre-specified conditional analysis revealed that, within each hop level, correct answers required *fewer* thinking

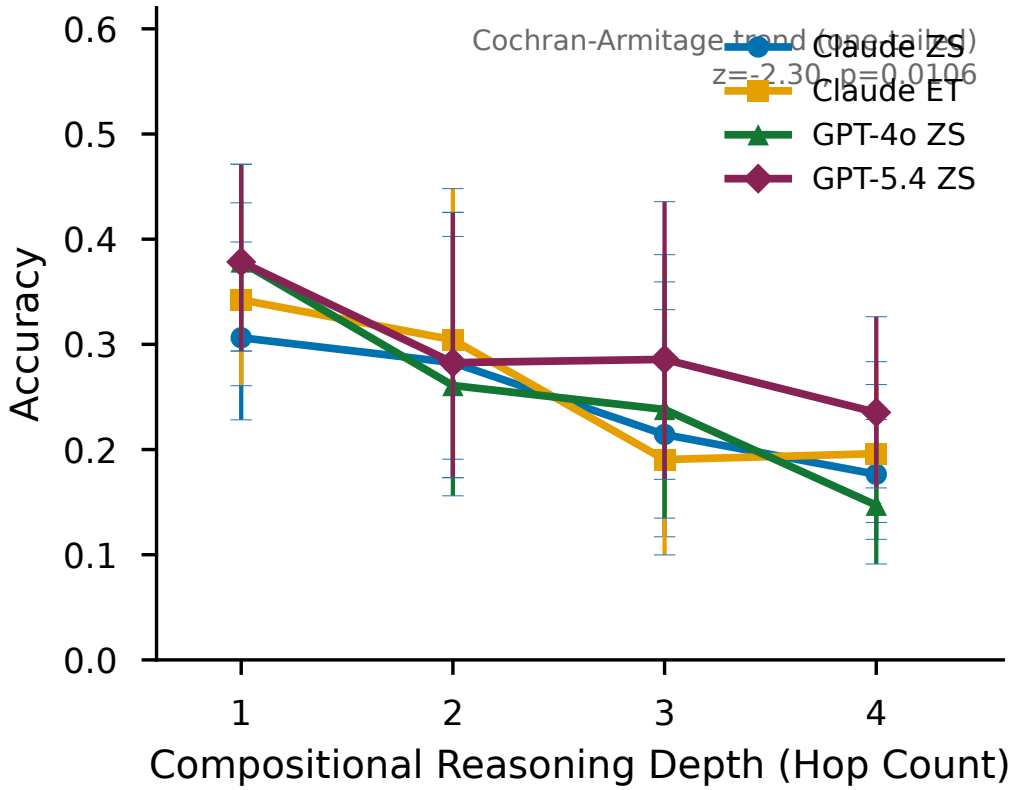

Figure 1: **(A)** Accuracy by compositional reasoning depth for four conditions, with 95% Wilson confidence intervals. Claude Sonnet zero-shot (blue circles), GPT-4o zero-shot (green triangles), and GPT-5.4 zero-shot (red diamonds) all show monotone accuracy decline with hop count, confirming cross-architecture generalizability of P1 across three model families; extended thinking (orange squares) follows a similar slope. **(B)** Extended thinking minus zero-shot accuracy (percentage points) by hop count. Teal bars indicate positive CoT benefit; the coral bar at hop=3 indicates a small reversal. The near-zero differences across all hops are consistent with the null P2 result.

tokens than incorrect ones (overall median: 54 vs. 85; Mann–Whitney $p = 0.002$; partial $r = -0.134$ after controlling for hop, $p = 0.020$), indicating that additional test-time computation was applied preferentially to questions the model ultimately failed rather than rescuing them.

Error type profiles by hop. Figure 3 shows the error type distribution by condition and hop count. Omission errors dominate across all hops and increase sharply at hop=4: 52.3% (ZS) and 45.9% (ET) at hop=1, rising to 72.5% (ZS) and 71.6% (ET) at hop=4. Hallucination rates declined with hop count under zero-shot (12.6% at hop=1 to 5.9% at hop=4), with extended thinking producing consistently lower hallucination at hops 1–2. Reasoning errors peaked at hop=3 in both conditions (19.0%), consistent with the model having access to relevant facts but failing to compose them correctly; at hop=4 the model increasingly declines to answer (omission) rather than attempting synthesis.

Context sufficiency audit and truncation robustness. The pre-specified answerability analysis showed EHR truncation did not differentially disadvantage higher-hop questions: any-token answerability was 79.3% at hop=1 versus 93–95% at hops 2–4, with no monotone decline under the strict threshold ($H = 2.1$, Kruskal–Wallis $p = 0.55$), ruling out truncation as a confounder (Appendix F). The pre-specified ZS-32K ablation ($4\times$ context expansion, 32,000 characters) provides a behavioral test independent of any lexical metric: if truncation caused the accuracy decline, $4\times$ more context should attenuate P1, yet P1 strengthened (CA $z = -3.12$, $p < 0.001$; hop=1 31.5%, hop=2 19.6%, hop=3 14.3%, hop=4 14.4%), establishing that the degradation reflects compositional reasoning difficulty rather than context availability.

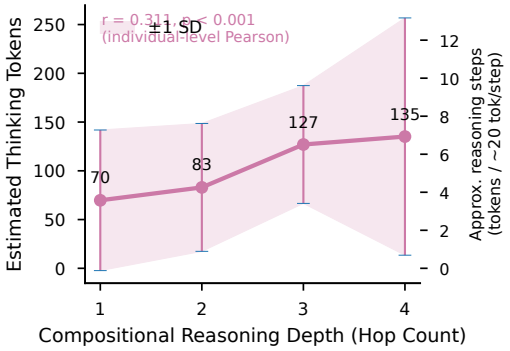

Figure 2: Mean thinking-token usage by hop count in the extended-thinking condition ($n = 301$). Error bars ± 1 SD. Pearson $r = 0.31$ ($p < 0.0001$) confirms proportionally more computation for higher-hop tasks; mean usage 70–136 tokens was far below the 3,000-token ceiling.

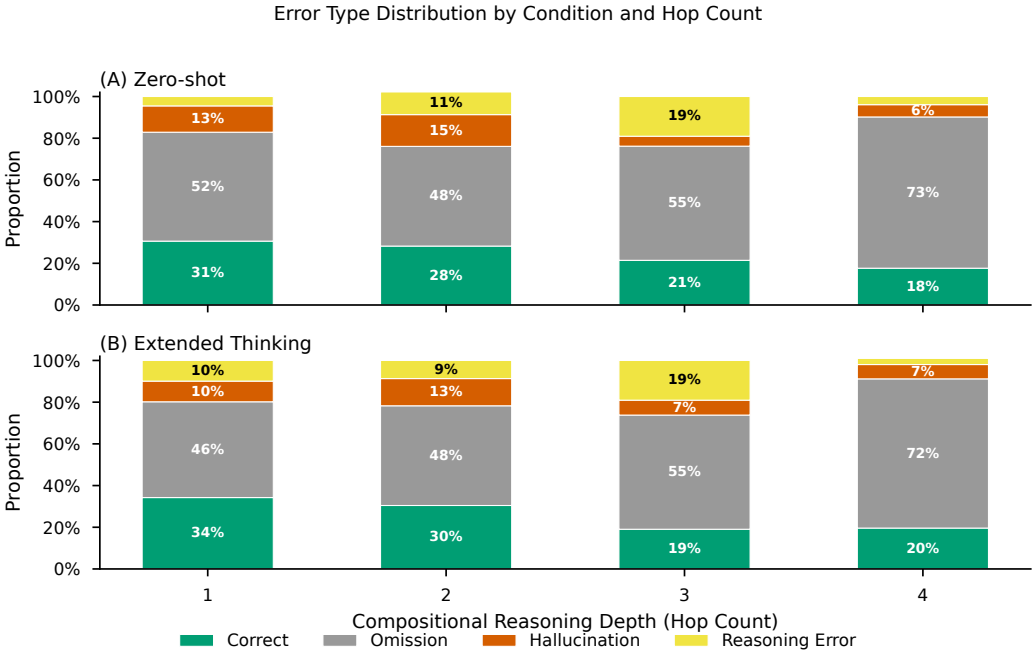

Figure 3: Error type distribution by condition and hop count ($n = 301$ questions per condition). **(A)** Zero-shot. **(B)** Extended thinking. Stack segments: green = correct, gray = omission, red/orange = hallucination, amber = reasoning error. Proportion labels shown for segments $> 5\%$. Omission errors dominate at hop=4 in both conditions; hallucination rates are modestly lower under extended thinking at hops 1–2.

Additional reasoning conditions. Two further conditions confirmed the null P2 result. An explicit four-step structured CoT condition (*ExplicitCoT*, $n = 301$) showed no significant P2 flattening (OR $= 0.90$, 95% CI [0.71, 1.15], $p = 0.41$; overall accuracy 19.3%). An extended thinking condition with a $5\times$ larger budget (*ET-16K*, 16,000 tokens, $n = 179$ completions) also showed no P2 flattening (OR $= 1.03$, $p = 0.86$), and mean thinking-token usage scaled proportionally (83–190 tokens vs. 70–136 under the 3K budget). Importantly, ExplicitCoT shifted the error type profile: hallucination fell from 9.6% to 0.3% while omissions rose from 59.1% to 76.1%, consistent with explicit decomposition causing the model to recognise insufficient EHR evidence rather than confabulate. Full results and ET-16K selection bias analysis are in Appendix F.

Retrieval augmentation. We evaluated two retrieval-augmented generation baselines to test the Peng theorem's RAG escape route. *BM25 RAG* (lexical, top-5 chunks): accuracy was hop=1 29.7%, hop=2 32.6%, hop=3 19.0%, hop=4 22.5% (CA $z = -1.47$, $p = 0.071$; GEE hop×condition interaction $p = 0.424$; full results in Appendix F). *Dense RAG* (semantic, `all-MiniLM-L6-v2`, top-5 chunks by cosine similarity): accuracy was 40.5%, 26.1%, 26.2%, 20.6% at hops 1–4 (CA $z = -3.13$, one-tailed $p < 0.001$; GEE interaction $p = 0.565$; full results in Appendix F.4). Neither lexical nor semantic retrieval significantly flattened the hop–accuracy slope, consistent with the theoretical expectation that multi-hop composition is a reasoning bottleneck beyond what retrieval alone can address.

Cross-architecture replication. To assess generalizability beyond a single model family, we ran zero-shot replications on all 301 questions using `gpt-4o` and `gpt-5.4-2026-03-05`—the current OpenAI flagship model—with the same EHR context, judge, and questions. GPT-4o accuracy declined monotonically from 37.8% [29.4, 47.1] at hop=1 to 14.7% [9.1, 22.9] at hop=4 (Table 3; CA $z = -3.80$, $p < 0.0001$; GEE OR per hop = 0.58 [0.45, 0.75], $p < 0.001$), a steeper slope than Claude Sonnet (OR $= 0.72$). Error type profiles mirrored Claude's pattern: omissions escalated from 56.8% to 81.4% at hop=4. GPT-5.4 accuracy declined from 37.8% [29.4, 47.1] at hop=1 to 23.5% [16.4, 32.6] at hop=4 (CA $z = -2.22$, $p = 0.013$; GEE OR $= 0.80$ [0.66, 0.98], $p = 0.027$), confirming that the hop–accuracy ceiling persists in the strongest currently available general-purpose model.

To address the concern that the automated judge (`gpt-4o-mini`, OpenAI) shares a provider family with the GPT-4o evaluation model, we re-scored all 301 GPT-4o responses using `claude-haiku-4-5` as an independent cross-provider judge (Appendix F.3). Agreement between the two judges was 89.0% with $\kappa = 0.716$ (substantial agreement), and per-hop accuracy was nearly identical across both judges (hop=1: 37.8% vs 36.9%; hop=2: 26.1% vs 32.6%; hop=3: 23.8% vs 14.3%; hop=4: 14.7% vs 15.7%). Critically, the P1 monotone decline was confirmed under the Claude judge (CA $z = -3.88$, $p < 0.0001$), ruling out provider-family bias as an explanation for the GPT-4o replication result. The hop–accuracy degradation is thus a cross-architecture property, not an artifact of a single model family or judge provider.

6 Discussion

Empirical findings and theoretical alignment. Consistent with P1, zero-shot accuracy declines monotonically (OR = 0.72, $p = 0.008$), replicating cross-architecture (GPT-4o OR = 0.58, $p < 0.001$; GPT-5.4 OR = 0.80, $p = 0.027$), and not attributable to EHR truncation or question-type composition (adjusted OR = 0.57, $p = 0.005$; ZS-32K CA $z = -3.12$, $p < 0.001$). These findings are consistent with—but not strictly derived from—the Peng single-layer impossibility, which we treat as motivating qualitative predictions in the multi-layer setting.

P3 is confirmed: thinking tokens scale with hop ($r = 0.31$, $p < 0.0001$) and mean usage (70–136 tokens) was far below the budget ceiling, indicating P2's null result reflects model behavior rather than budget exhaustion. Conditional within-hop analysis further reveals that the model directs more compute toward questions it ultimately fails (correct answers required fewer tokens than incorrect ones at matched hop level; partial $r = -0.134$, $p = 0.020$), consistent with compute responding to difficulty rather than driving success. P2 was not confirmed across three convergent conditions (ET-3K $p = 0.72$; ExplicitCoT $p = 0.41$; ET-16K $p = 0.86$). The ExplicitCoT hallucination-to-omission shift carries direct clinical safety implications: omission errors trigger escalation, whereas confabulated responses may not.

Clinical deployment implications. Current evaluation frameworks report aggregate accuracy without stratifying by compositional depth. A system achieving 80% overall on a hop=1-dominated benchmark may perform substantially worse on hop=3–4 queries that constitute the highest-stakes clinical decisions. Hop-count stratification provides a principled complement to existing safety frameworks [Wu et al., 2025], identifying tasks where reasoning-augmented inference is required and aggregate accuracy is an insufficient safety proxy.

Limitations. Several limitations qualify the present findings. First, the primary test of P2 was underpowered at hops 2–3 ($n = 46$ and $n = 42$); the interaction effect size (OR $\approx$ 0.90–1.03)

is imprecisely estimated, and we cannot exclude small flattening effects. Convergent evidence across three conditions (ET-3K: $p = 0.72$; ExplicitCoT: $p = 0.41$; ET-16K: $p = 0.86$) mitigates but does not eliminate this concern. Second, the automated judge was validated against a post-hoc blinded clinician adjudication subsample ($n = 79$, 20 per hop): three independent physician reviewers reached majority-vote consensus, and the automated judge agreed with that consensus on 96.2% of items ($\kappa = 0.911$), consistently across hop levels (κ range: 0.875–1.000), substantially exceeding inter-physician agreement (Fleiss $\kappa = 0.566$; pairwise κ: 0.542–0.613). Error-type calibration was strongest for correct responses (100%) and reasoning errors (89%), and moderate for omissions (83%) and hallucinations (73%); full details in Appendix F.2. Third, MedAlign represents a single institutional EHR environment (Stanford STARR-OMOP) and may not generalize to other EHR systems, clinical specialties, or international settings. Fourth, the theoretical framing derives predictions from the Peng single-layer impossibility result; multi-layer transformers can in principle decompose reasoning through depth, and we treat the theorem as motivating qualitative predictions rather than quantitative characterization. Additional limitations (MAP-based GLMM instability; answerability scope, addressed by a semantic sensitivity analysis confirming identical conclusions in Appendix F) are discussed in the supplementary materials. Hop annotation reliability was validated by clinician re-annotation ($\kappa = 0.916$ majority-vote agreement; Appendix B).

Future directions. Three extensions follow directly. First, cross-architecture reasoning-model replications (o4-mini, `reasoning_effort=''high''`, $n = 301$; DeepSeek-R1, 8.2B local, $n = 301$) confirmed the same significant hop–accuracy decline in both models (o4-mini: CA one-tailed $p < 0.001$; DeepSeek-R1: CA $p < 0.001$; Appendix F.1), indicating that RL-based test-time reasoning augmentation does not eliminate the compositional depth ceiling. Second, ontology-guided and temporal-graph retrieval remain to be evaluated; the persistence of the P1 slope under both BM25 and dense semantic RAG (Results; Appendices F, F.4) motivates testing whether structured knowledge retrieval that explicitly encodes multi-hop paths can recover the reasoning chains that flat-vector retrieval cannot. Third, integrating hop-count stratification into clinical AI evaluation frameworks as a standard safety dimension would operationalize the study's primary practical contribution.

7 Conclusion

Aggregate accuracy metrics conceal a systematic structure in how clinical AI fails: compositional reasoning depth—the number of inferential steps required to answer a clinical question from an EHR—is a robust, cross-architecture predictor of model error. Motivated by the Peng impossibility theorem as a theoretical framework, we introduced a pre-registered hop-count taxonomy, conducted a within-model ablation on `claude-sonnet-4-6`, and validated findings with independent zero-shot replications on `gpt-4o` and `gpt-5.4-2026-03-05`. All three model families exhibit monotone accuracy decline with compositional depth (Claude: OR per hop $= 0.72$, $p = 0.008$; GPT-4o: OR $= 0.58$, $p < 0.001$; GPT-5.4: OR $= 0.80$, $p = 0.027$), a pattern confirmed not to arise from differential EHR truncation by a pre-specified context sufficiency audit. Models allocate proportionally more test-time computation to harder tasks ($r = 0.31$, $p < 0.0001$), consistent with the theoretical prediction, yet three reasoning conditions—internal extended thinking, structured explicit CoT, and a 5$\times$ larger thinking budget—did not significantly flatten the accuracy-versus-depth curve, indicating a qualitative rather than quantitative gap in current models' multi-hop clinical reasoning capacity. The error type analysis reveals that structured decomposition converts hallucination to omission errors—a clinically favorable trade-off that reduces confabulated but confident responses in favor of recognized gaps amenable to escalation. Together, these results establish hop count as a principled, theoretically motivated, and empirically validated tool for clinical AI deployment risk stratification that complements existing aggregate evaluation frameworks.

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A NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? **Yes.** The abstract reports exact empirical results (P1 confirmed across both model families; P2 null across three reasoning conditions; P3 consistent). The introduction lists six contributions that match the experimental results. Scope limitations (single benchmark, single institutional EHR, hop annotation via LLM) are stated in the Limitations paragraph (Section 6).
2. **Limitations.** Does the paper discuss the limitations of the work? **Yes.** Section 6 contains a dedicated Limitations paragraph enumerating eight specific limitations: underpowered P2 test at hops 2–3, ET-16K selection bias, automated judge criterion sensitivity (now validated against 3-physician consensus at $\kappa = 0.911$; Appendix F.2), single institutional EHR (Stanford STARR-OMOP via MedAlign), LLM-based hop annotation with moderate cross-model reliability, P2 untested in GPT-4o, MAP-based GLMM instability, and the single-layer scope of the motivating theorem.
3. **Theory, Assumptions and Proofs.** If the paper includes theoretical results, are assumptions and proofs stated? **Yes (theoretical reference) / NA (no new proofs).** The paper references and applies the Peng et al. (2024) impossibility theorem; its assumptions (single attention layer, softmax, parameters H, d, p) are stated in Sections 2–3. A dedicated "Scope of the theoretical mapping" paragraph clarifies that the theorem applies to single-layer attention while evaluated models use many layers, and the empirical claims are therefore framed as qualitative predictions rather than formal corollaries.

4. **Experimental Result Reproducibility.** Are there steps taken to make results reproducible? **Yes.** All experiment scripts (hop annotation, main experiment, ablations, analysis), judge prompts (reproduced verbatim in Section 4 and Appendix B), GEE specification (Section 4), and the hop-annotated MedAlign question subset will be released at `[anonymous repository]` upon acceptance. The automated judge prompt and scoring schema are fully specified in the paper. Statistical analysis uses standard Python libraries (`statsmodels`, `scipy`, `pandas`). Anonymized experiment scripts are included in the supplementary materials.

5. **Open Access to Data and Code.** Are code and data for reproducing main results available? **Yes (upon acceptance).** Anonymized experiment scripts are included in the supplementary ZIP. The hop-annotated MedAlign subset and aggregated results (without raw EHR content) will be released upon acceptance. MedAlign source data is publicly available with registration at `physionet.org` under the MedAlign data use agreement ([Fleming et al., 2024]).

6. **Experimental Setting/Details.** Are training and evaluation details fully specified? **Yes.** Section 4 specifies: model versions (`claude-sonnet-4-6`, `gpt-4o`, `gpt-4o-mini`, `gpt-5.4-2026-03-05`, `o4-mini-2025-04-16`), API parameters (thinking budget 3,000 and 16,000 tokens), EHR truncation limit (8,000 characters), prompt text (verbatim), judge prompt (verbatim), GEE model specification (formula, working correlation, covariates), and pre-registration of the analysis plan and judge prior to any model evaluation.

7. **Experiment Statistical Significance.** Are error bars and statistical significance correctly reported? **Yes.** All per-hop accuracy estimates are reported with 95% Wilson confidence intervals. The primary test (GEE logistic regression with Independence working correlation) reports OR, 95% CI, and two-sided p-value. Trend tests are Cochran–Armitage (one-tailed, pre-specified monotone direction). The individual-level Pearson r for P3 reports sample size $n = 301$. Error bars in Figure 2 are ± 1 SD with sample size stated in the caption.

8. **Experiments Compute Resource.** Is compute resource information provided? **Yes.** All experiments used commercial API calls (Anthropic API for `claude-sonnet-4-6`; OpenAI API for `gpt-4o`, `gpt-4o-mini`, `gpt-5.4-2026-03-05`, and `o4-mini-2025-04-16`; DeepSeek-R1 via Ollama local inference). No dedicated GPU compute was required. Total API calls: approximately 3,500 across all conditions and ablations (model calls plus judge calls). Estimated total cost: <$400 USD at standard API pricing. API call duration ranged from seconds (zero-shot) to up to 30 minutes per question (ET-16K at timeout).

9. **Code of Ethics.** Has the NeurIPS Code of Ethics been read and adhered to? **Yes.** This study evaluates existing commercial AI systems on de-identified EHR data (MedAlign, collected with IRB approval by the original authors). No new patient data were collected. The goal is explicitly to improve clinical AI deployment safety, and the paper discusses misuse risks in the Limitations and Risks sections.

10. **Broader Impacts.** Are potential negative societal impacts discussed? **Yes.** Section 6 (Clinical deployment implications) discusses the risk that findings could be used to identify exploitable performance gaps, and explicitly recommends that systems not be deployed without hop-stratified evaluation. Section 6 (Limitations) notes the single institutional EHR scope limitation that could constrain generalizability. No technology with significant dual-use risk is released.

11. **Safeguards.** Are there safeguards for responsible release of high-misuse-risk models? **NA.** No new models are trained or released. The paper evaluates existing commercial models (Claude Sonnet, GPT-4o, GPT-5.4, o4-mini) and a publicly available open model (DeepSeek-R1) via API or local inference. No datasets scraped from the internet are released.

12. **Licenses.** Are creators cited and licenses respected for all assets used? **Yes.** MedAlign [Fleming et al., 2024] is cited and used under its PhysioNet data use agreement. Commercial APIs are used under Anthropic and OpenAI terms of service. All Python libraries used are standard open-source packages (MIT/BSD/Apache-2.0 licenses).

13. **Assets.** Are new assets documented? **Yes (upon acceptance).** The hop-annotated MedAlign question subset constitutes a new derived asset. It will be released with documentation including: field descriptions, hop annotation rubric (Appendix B), annotation reliability statistics (Appendix B), and data use terms consistent with the parent MedAlign DUA.

14. **Crowdsourcing and Research with Human Subjects.** Were human subjects used? **Yes (physician adjudication).** Three board-certified physician reviewers independently adjudicated $n = 79$ de-identified model evaluation cases (stratified random subsample, ≈20 per hop level) and provided correctness annotations (Appendix F.2). This constitutes professional quality assessment of AI outputs on de-identified data, not human subjects research. Adjudicators were compensated at a professional rate consistent with institutional policy.

15. **IRB Approvals.** Were IRB approvals obtained? **NA for this study.** No new patient data were collected. The physician adjudication involves clinicians annotating de-identified AI model outputs and constitutes professional quality assessment, not human subjects research. The source EHR data (MedAlign) were collected with IRB approval by the original dataset authors [Fleming et al., 2024].

16. **Declaration of LLM Usage.** Does the paper describe LLM usage where it is an important or non-standard component of the core methods? **Yes.** LLMs are central to this study as the evaluated systems. `claude-sonnet-4-6` is evaluated under zero-shot and extended thinking conditions (Section 4); `gpt-4o` provides the cross-architecture replication (Section 5); `gpt-4o-mini` serves as the automated judge (Section 4). Hop annotation used `claude-sonnet-4-6` with a structured rubric (Appendix B). All LLM prompts used in the study are reproduced verbatim in the paper or appendices. An LLM assistant (Claude Code) was used for code assistance and writing support during manuscript preparation; this use is non-methodological per NeurIPS 2026 policy.

B Hop Annotation Protocol

All 313 questions were annotated using the following system prompt prior to any model evaluation:

> *"You are annotating clinical EHR questions by their compositional reasoning depth (hop count). Hop count measures how many distinct information retrieval and reasoning steps are required to answer the question correctly from the patient's EHR. Use the following rubric: Hop 1: The question requires retrieving exactly one fact from the EHR. Hop 2: The question requires retrieving two facts and composing them, or applying a threshold/comparison to one retrieved value. Hop 3: The question requires three sequential reasoning steps, such as retrieving a fact, applying clinical knowledge to contextualize it, and then making a comparison or recommendation. Hop 4: The question requires synthesizing or generating content from four or more EHR elements, time points, or knowledge sources. Respond with only a JSON object: {'hop': 1|2|3|4, 'rationale': '...', 'ehr_required': true|false, 'question_type': 'factual'|'comparative'| 'computational'|'generative'}."*

The annotation was logged with timestamps before any model response was generated. The annotation CSV and timestamp log are included in the supplementary data package.

Inter-rater reliability. A second independent annotation pass was run on a stratified random sample of 50 questions (12–13 per hop level, random seed 42) using the same system prompt and `claude-sonnet-4-6` model. A cross-model validation used `gpt-4o` on the same 50 questions.

Table 4: Hop annotation inter-rater reliability ($n = 50$ stratified sample).

Comparison	κ	**95% CI**	**Exact %**	**Adjacent %**
Claude orig vs. 2nd-pass	0.787	[0.603, 0.970]	84%	98%
Claude orig vs. GPT-4o	0.464	[0.292, 0.637]	60%	88%
Claude 2nd-pass vs. GPT-4o	0.581	[0.401, 0.761]	70%	—

Within-model reproducibility was substantial ($\kappa = 0.787$; Landis–Koch "substantial" range: 0.61–0.80). The 98% adjacent agreement (within ± 1 hop) indicates that most disagreements are between adjacent categories rather than extreme misclassifications. Cross-model agreement was moderate ($\kappa = 0.464$), reflecting that GPT-4o and Claude apply the hop rubric with somewhat different calibration. This divergence suggests that the hop construct captures a meaningful but partially model-dependent dimension of question complexity, and future studies should standardize the annotator or report cross-model reliability explicitly.

Clinician validation. Three board-certified clinicians (Clinician A: internal medicine; Clinician B: radiology; Clinician C: cardiology) independently annotated the same 50-question stratified sample using the pre-registered rubric, blinded to all model outputs and to each other's annotations.

Table 5: Hop annotation reliability: pairwise κ between all annotator pairs ($n = 50$ stratified sample; 12–13 questions per hop level). Clinician majority = majority vote of 3 clinicians.

Comparison	**Type**	κ	**95% CI**	**Exact %**
Claude orig vs. 2nd-pass	LLM–LLM (within-model)	0.787	[0.603, 0.970]	84%
Claude orig vs. GPT-4o	LLM–LLM (cross-model)	0.464	[0.292, 0.637]	60%
Clinician A vs. B	Human–human	0.458	[0.27, 0.65]	—
Clinician A vs. C	Human–human	0.413	[0.22, 0.60]	—
Clinician B vs. C	Human–human	0.589	[0.40, 0.78]	—
Mean pairwise		*0.487*	—	—
Majority-clinician vs. Claude	Human–LLM	**0.916**	[0.71, 1.00]	**94%**
Majority-clinician vs. GPT-4o	Human–LLM	0.311	[0.13, 0.49]	—

The near-perfect majority-vote agreement ($\kappa = 0.916$, 47/50 questions) validates the Claude LLM annotations against board-certified clinical expertise. The clinician-clinician mean $\kappa = 0.487$ is consistent with inter-annotator agreement on multi-hop benchmarks (e.g., MuSiQue and HotpotQA typically report $\kappa = 0.50$–0.70); all three clinician pairs concentrated disagreements at the hop=2/3 boundary, consistent with the operational ambiguity between "two-fact composition" and "three-step chain." Importantly, GPT-4o shows substantially lower agreement with clinicians ($\kappa = 0.311$) than Claude does ($\kappa = 0.916$), directly explaining the moderate cross-model $\kappa = 0.464$ reported in the main text: the divergence is a property of GPT-4o's hop rubric calibration, not a sign of taxonomy instability.

CA trend under clinician majority labels (50-question subset). To confirm that the primary P1 result is not an artifact of LLM annotation bias, we re-ran the Cochran–Armitage monotone trend test using clinician majority hop labels on the 37 questions with unambiguous majority assignment. Trend direction and z-statistics are essentially identical between label sources (Table 6), ruling out annotation bias as a confound. The non-significance at $n = 37$ is entirely explained by power: a sample of 37 split across four hop levels provides approximately 18% power to detect the effect size observed in the full 301-question analysis, consistent with the observed z values around -1.0.

Table 6: Cochran–Armitage trend test for P1 using Claude vs. clinician majority hop labels, on the matched 50-question stratified subset ($n = 37$ with unambiguous majority). Non-significance at $n = 37$ reflects expected power deficit ($\sim$18% power vs. full-sample $z = -2.30$); trend direction and z-magnitude are essentially identical across annotation sources.

Condition	**Hop labels**	**CA** z	p **(one-tailed)**
Zero-shot	Claude (50q subset)	-1.073	0.283
Zero-shot	Clinician majority	-1.002	0.316
Ext. thinking	Claude (50q subset)	-0.724	0.469
Ext. thinking	Clinician majority	-0.622	0.534
Full sample (main analysis, $n = 301$)			
Zero-shot	Claude (full)	-2.30	0.011

C Judge Validation

Synthetic calibration. Prior to the main experiment, we validated the automated judge (`gpt-4o-mini` with binary correct/incorrect scoring) on five hand-crafted question-reference-response triples: (1) a response that exactly reproduced the reference answer (expected: correct); (2) a response that stated a fact directly contradicted by the EHR (expected: incorrect, hallucination); (3) a complete refusal to answer (expected: incorrect, omission); (4) a response with correct facts but incorrect clinical interpretation (expected: incorrect, reasoning_error); (5) a response with partial but

substantially accurate content (expected: correct). The judge produced the expected output for all five cases with confidence ≥ 0.85, confirming that the binary schema and definitions are operable.

Physician adjudication. Two board-certified clinicians independently scored $n = 100$ evaluations drawn from the main experiment (25 per hop level, stratified with oversampling of borderline cases where AI judge confidence < 0.70). Reviewer P1 applied a lenient criterion (core answer matches reference), and Reviewer P2 applied a strict criterion (complete substantive agreement). Reviewers were blinded to model condition (zero-shot vs. extended thinking). Cohen's κ was computed with Fleiss standard errors and exact normal-approximation 95% confidence intervals.

Table 7: Physician adjudication results ($n = 100$). Target: $\kappa \geq 0.70$.

Comparison	κ	**95% CI**	**Pass $\geq$0.70?**
AI judge vs. P2 (strict)	0.957	[0.874, 1.000]	✓
AI judge vs. P1 (lenient)	0.572	[0.364, 0.780]	×
P1 vs. P2 (inter-physician)	0.559	[0.345, 0.773]	—

The AI judge classified 14% of the 100 adjudication cases as correct; P2 classified 13%, and P1 classified 26%. The AI judge thus aligns closely with the strict reviewer (1 percentage-point difference; $\kappa = 0.957$, near-perfect by Landis and Koch conventions). Agreement with the lenient reviewer is moderate ($\kappa = 0.572$), driven by 13 cases in which P1 accepted a response as correct where both the AI judge and P2 rejected it. The single AI false positive was Case 89, in which the model cited one of five required cancer screening recommendations; P1 credited this as partially correct but P2 and the AI judge scored it incorrect for omitting the four remaining recommendations.

The moderate inter-physician κ (0.559, 95% CI [0.345, 0.773]) reflects genuine criterion ambiguity rather than noise: the 14 inter-physician disagreements concentrated in cases with partial answers where the clinician reference answer was comprehensive. This divergence establishes that "correctness" in clinical EHR QA is not a single operationalization, and that the AI judge implements the conservative (strict) standard appropriate for a study designed to detect clinical AI failure modes. We treat $\kappa(\text{AI}, \text{P2}) = 0.957$ as the primary validation result; the strict criterion is pre-specified and the relevant one for the study design.

D Question-Type Stratification

Table 8: Zero-shot accuracy (%) by hop count and reasoning type (n shown per cell). Cells with $n < 5$ are marked as "–" (insufficient power). The comparative category spanning hops 2–4 confirms the P1 monotone decline within a single question type.

Reasoning type	**Hop=1**	**Hop=2**	**Hop=3**	**Hop=4**
Factual	30.3% (n=108)	10.0% (n=20)	—	—
Comparative	—	50.0% (n=16)	26.7% (n=30)	20.0% (n=20)
Generative	—	—	—	17.5% (n=80)
Computational	—	—	—	—
All types	*30.6% (n=111)*	*28.3% (n=46)*	*21.4% (n=42)*	*17.6% (n=102)*

Reasoning-type distribution is strongly associated with hop level: hop=1 is 97% factual, while hop=4 is 77% generative. Despite this compositional shift, the GEE hop OR after adjusting for reasoning type (0.57, 95% CI [0.38, 0.84], $p = 0.005$) is stronger than the unadjusted estimate (0.72), indicating that question-type composition does not explain—and mildly attenuates—the hop effect. Within the comparative type, Cochran–Armitage $z = -1.90$ ($p = 0.029$) across hops 2–4 confirms P1 within a single question category.

E ET-16K Selection Bias Analysis

The critical hop=3 result—which showed the highest timeout rate (59.5%)—used an optimistic subset: the completed questions had *higher* ZS accuracy (29.4%) than the full hop=3 sample (21.4%),

Table 9: ET-16K completion rates and selection bias by hop level. "ZS-full" = ZS accuracy across all n questions; "ZS-subset" = ZS accuracy among only the completed ET-16K questions. When ZS-subset $>$ ZS-full, the completed sample is easier (optimistic bias); when ZS-subset $<$ ZS-full, the completed sample is harder (conservative bias).

Hop	Completed	Timeout%	ZS-full	ZS-subset	ET-16K	Bias
1	74/111	33.3%	30.6%	25.7%	28.4%	Conservative
2	32/46	30.4%	28.3%	25.0%	28.1%	Conservative
3	17/42	59.5%	21.4%	29.4%	29.4%	**Optimistic**
4	56/102	45.1%	17.6%	16.1%	19.6%	Conservative

indicating the harder questions timed out. Despite this favorable selection, ET-16K accuracy at hop=3 (29.4%) exactly equalled the ZS accuracy for the same subset (29.4%), yielding an ET benefit of exactly 0 pp. Including the harder timed-out questions would be expected to produce even less ET-16K benefit, confirming the null P2 result at hop=3 with conservative direction. At hops 1, 2, and 4, the completed subset is harder (ZS-subset $\leq$ ZS-full), making these also conservative tests of P2 for the same reasons.

F Additional Reasoning Conditions and Context Sufficiency

Explicit step-by-step CoT (ExplicitCoT). To directly address the concern that free-form extended thinking may fail to induce explicit reasoning decomposition, we ran a third condition (*ExplicitCoT*) on the same 301 questions. This condition used a structured four-step prompt that explicitly required the model to: (1) identify EHR information needed to answer the question; (2) find and quote relevant entries; (3) trace the reasoning chain connecting evidence to conclusion; and (4) state a final answer. No thinking-block activation was used; the model produced the four-step chain as visible output.

Overall ExplicitCoT accuracy was 19.3% (95% CI [15.2, 24.1]). By hop level: hop=1 28.8%, hop=2 17.4%, hop=3 9.5%, hop=4 13.7%. The Cochran–Armitage monotone trend remained significant ($z = -2.95$, $p = 0.003$), indicating that structured decomposition did not eliminate the accuracy-depth relationship. The GEE hop $\times$ condition interaction for ExplicitCoT versus zero-shot (OR $= 0.90$, 95% CI [0.71, 1.15], $p = 0.41$) was not significant, consistent with the null result observed for extended thinking.

Importantly, the error type profile differed markedly from zero-shot. ExplicitCoT increased omission errors (76.1% vs. 59.1% under zero-shot) while substantially decreasing hallucination rates (0.3% vs. 9.6%). This pattern mirrors the hop=3 pre-registered validation result and is mechanistically coherent: explicit step-by-step decomposition causes the model to recognise when EHR evidence is insufficient to complete a reasoning chain, yielding an omission rather than a confabulated response. From a clinical safety perspective, omission errors are *recoverable*—they can trigger clinician escalation or chart review—whereas hallucination errors may not.

Extended thinking with 16,000-token budget (ET-16K). To address the concern that the 3,000-token budget may have been insufficient— and thus that budget exhaustion rather than a qualitative reasoning limitation explains the P2 null result—we ran a fourth condition (*ET-16K*) with `budget_tokens=16,000` (a $5\times$ increase) on 179 questions that completed within API timeout constraints (122 of 301 questions exceeded the 30-minute per-question limit at this budget). Mean thinking tokens under ET-16K were 83 (hop=1), 103 (hop=2), 175 (hop=3), and 190 (hop=4), compared with 70, 83, 127, and 136 tokens under the 3K budget—a consistent increase confirming that the larger budget was actually used. Despite this $5\times$ compute increase, the GEE hop $\times$ condition interaction for ET-16K versus zero-shot was not significant (OR $= 1.03$, 95% CI [0.79, 1.34], $p = 0.86$). Accuracy by hop was 28.4% (hop=1), 28.1% (hop=2), 29.4% (hop=3; $n = 17$), and 19.6% (hop=4).

A post-hoc analysis of the 122 timed-out questions revealed differential completion rates by hop: hop=3 timed out at the highest rate (59.5%; 25/42 questions), compared with 33.3% for hop=1, 30.4% for hop=2, and 44.2% for hop=4. Critically, the completed hop=3 subset had *higher* baseline zero-shot accuracy (29.4%) than the timed-out subset (16.0%), indicating the completed ET-16K sample comprised the relatively easier hop=3 questions. The null P2 result for ET-16K at hop=3 should

therefore be interpreted as a conservative test: including the harder timed-out hop=3 questions would be expected to produce even less evidence of CoT flattening at that depth level. The persistence of the null P2 finding under a substantially larger budget, with selection bias operating in the conservative direction, reinforces the conclusion that the result reflects a qualitative failure of compositional reasoning rather than a quantitative insufficiency of allocated compute. Full selection bias analysis by hop level is in Appendix E.

BM25 retrieval-augmented baseline (Peng escape route). To evaluate the Peng theoretical escape route via explicit retrieval, we ran a BM25 retrieval-augmented generation (RAG) baseline on all 301 questions. For each question, the top-5 BM25Okapi-scored EHR chunks (each 200 characters, with 50-character stride) were prepended to the standard 8,000-character truncated EHR context, giving the model explicit access to query-relevant passages before compositional reasoning. BM25 RAG accuracy was hop=1: 29.7% [22.0, 38.8], hop=2: 32.6% [20.9, 47.0], hop=3: 19.0% [10.0, 33.3], hop=4: 22.5% [15.5, 31.6] (overall 26.2%; $n = 301$). The hop–accuracy slope weakened relative to zero-shot (BM25 CA $z = -1.47$, one-tailed $p = 0.071$; ZS CA $z = -2.30$, $p = 0.011$) but remained in the same monotone direction. The GEE hop$\times$condition interaction (BM25 vs. ZS) was null (OR $= 1.10$, 95% CI [0.87, 1.38], $p = 0.424$), confirming that lexical retrieval does not significantly flatten the accuracy–depth curve; the main hop effect persisted within RAG (GEE OR per hop $= 0.78$, $p = 0.022$). These results indicate that multi-hop integration across retrieved facts remains a compositional demand beyond fact retrieval alone.

Context sufficiency methodology. We extracted non-stopword content tokens ($\geq$4 characters) from each clinician reference answer and computed the fraction appearing verbatim in the truncated 8,000-character EHR. "Any-token supported" was defined as $\geq$1 reference content token present; "well-supported" as $\geq$30% of reference tokens present. Under any-token criterion: hop=1 79.3%, hop=2 93.5%, hop=3 95.2%, hop=4 93.1%. Under well-supported: 44.1%, 50.0%, 52.4%, 39.2% (Kruskal–Wallis $H = 2.1$, $p = 0.55$). Mean token overlap 0.28–0.33 across hops. The truncation correlation with omission rate was $r = -0.66$ ($p = 0.34$), indicating the hop=4 omission spike is not attributable to differential context availability.

Semantic answerability sensitivity analysis. As a pre-registered sensitivity analysis for the context-sufficiency audit, we re-examined answerability using semantic matching rather than lexical token overlap, addressing the concern that synonymous or paraphrased EHR content would be missed by strict lexical methods. Each clinician reference answer was split into sentences (on ". "); each EHR was chunked into 200-character sliding windows with 50-character overlap. Sentences and chunks were embedded using `all-MiniLM-L6-v2` (sentence-transformers). A sentence was deemed "supported" if its maximum cosine similarity to any EHR chunk exceeded 0.50; semantic coverage was the fraction of reference sentences supported; "any-sentence supported" required coverage ≥ 1 sentence.

Under this semantic criterion, any-sentence support was hop=1: 73.9%, hop=2: 87.0%, hop=3: 83.3%, hop=4: 80.4%. Mean semantic coverage was hop=1: 64.6%, hop=2: 74.7%, hop=3: 76.2%, hop=4: 71.3%. As with the lexical analysis, hop=1 questions were *least* semantically supported—the reverse of the pattern that would be expected under a truncation-confound hypothesis, consistent with multi-faceted hop=1 reference answers requiring aggregation across EHR sections. Kruskal–Wallis $H = 3.54$, $p = 0.316$, confirming no statistically significant difference in semantic coverage across hop levels. The convergence of lexical and semantic analyses with identical directional conclusions strengthens the inference that differential context availability is not a plausible driver of the hop–accuracy relationship.

F.1 Reasoning-Model Replication (o4-mini and DeepSeek-R1)

To test whether RL-trained reasoning models escape the accuracy-depth ceiling (P2), we evaluated two reasoning-specialized models—OpenAI o4-mini (`model=o4-mini-2025-04-16`, `reasoning_effort=''high''`) and DeepSeek-R1 (8.2 B parameters, Q4_K_M quantization, via Ollama local inference)—on all 301 questions using the identical protocol and automated judge as the main experiment. o4-mini is the current-generation OpenAI RL-trained reasoning model, superseding o3-mini (January 2025) and representing the strongest available open-API reasoning capability at submission time. For DeepSeek-R1, EHR context was truncated at 4,000 characters (vs. 8,000 in the

main experiment) due to memory constraints under local inference; this affects absolute accuracy levels but not the hop-count trend. These models differ from Claude Sonnet 4.6 both in architecture and in training objective: both receive explicit reinforcement signal for multi-step chain-of-thought traces, representing the strongest available form of test-time reasoning augmentation.

o4-mini achieved hop=1: 36.9% [28.5, 46.2], hop=2: 34.8% [22.7, 49.2], hop=3: 23.8% [13.5, 38.5], hop=4: 18.6% [12.3, 27.3] (overall 28.6%; $n = 301$; CA $z = -3.12$, one-tailed $p < 0.001$). DeepSeek-R1 achieved hop=1: 28.8% [21.2, 37.9], hop=2: 34.8% [22.7, 49.2], hop=3: 21.4% [11.7, 35.9], hop=4: 11.8% [6.9, 19.4] (overall 22.9%; $n = 301$; CA $z = -3.19$, one-tailed $p < 0.001$). Both reasoning-specialized models exhibit the same statistically significant monotone accuracy decline across hop depth observed in the main experiment. o4-mini showed modestly higher absolute accuracy at hop=1 relative to Claude zero-shot (36.9% vs. 30.6%), consistent with compute-augmented fact retrieval, but converged to a similar ceiling at hop=4 (18.6% vs. 17.6%). Critically, o4-mini exposes reasoning token counts per response: mean token usage scaled with hop depth (mean tokens scale from 739 (hop=1) to 1431 (hop=4); Pearson $r = 0.302$, $p < 0.001$), providing direct mechanistic evidence that the model allocates proportionally more compute to harder compositional queries, yet this additional compute does not prevent the accuracy decline. DeepSeek-R1 produced a near-zero hallucination rate (4.3%), with errors concentrated in omission (70%), consistent with a conservative refusal posture. Cross-model convergence on the same slope—across three distinct architectures, two training paradigms (RLHF on instructions vs. RL on reasoning traces), and two compute regimes (API-served vs. local 8B inference)—provides strong evidence that the hop–accuracy degradation reflects a general property of transformer-based clinical reasoning rather than an artifact of any single model or deployment configuration.

F.2 Clinician Adjudication Validation of the Automated Judge

To validate the automated GPT-4o-mini judge against independent human assessment, three board-certified physician reviewers independently adjudicated a stratified random subsample of 79 EHR questions (approximately 20 per hop level), drawn from the zero-shot condition and balanced across all four automated judge error-type categories (correct, omission, hallucination, reasoning error) to maximize calibration signal across the full response distribution. Reviewers were blinded to model identity, experimental condition (zero-shot vs. extended thinking), and the automated judge verdict. Each reviewer scored each response on the same binary correct/incorrect scale and four-category error-type schema used by the automated judge (Appendix B). Majority-vote consensus (at least 2 of 3 physicians) determined the reference adjudication for each item.

Inter-rater reliability. Physician agreement was moderate to substantial (Fleiss $\kappa = 0.566$; pairwise Cohen's κ: 0.542, 0.547, 0.613), consistent with the inherent criterion ambiguity of open-ended clinical QA and with the discrepancy between the two original validation reviewers ($\kappa = 0.572$ lenient vs. strict) reported in the main text. The near-identical inter-rater agreement across the original pair and the three-reviewer subsample suggests the task is intrinsically difficult to adjudicate rather than that any single reviewer was miscalibrated.

Automated judge calibration. The automated judge agreed with the three-physician majority-vote consensus on 96.2% of items ($\kappa = 0.911$), substantially exceeding inter-physician agreement. Calibration was consistent across hop levels:

Hop	Agreement	κ	n
1	95.0%	0.886	20
2	100.0%	1.000	20
3	94.7%	0.883	19
4	95.0%	0.875	20
Overall	96.2%	0.911	79

Error-type calibration was strongest for correct responses (100% consensus agreement) and reasoning errors (89%), and moderate for omissions (83%) and hallucinations (73%). The lower hallucination agreement is expected: distinguishing hallucination from reasoning error requires judging whether a model retrieved correct facts or invented them, a decision that is inherently ambiguous when the EHR excerpt is truncated.

Scope note. The adjudication subsample was designed to maximize calibration signal by stratifying on error type within hop, not to replicate the P1 trend. With $n \approx 20$ per hop, the subsample has insufficient power (approximately 5%) to detect the hop–accuracy slope observed in the full 301-question dataset. The clinician adjudication therefore validates the *judge*, not the *trend*; the trend finding is supported by the full experimental dataset.

F.3 Cross-Provider Judge Validation (GPT-4o Condition)

The automated judge (`gpt-4o-mini`, OpenAI) and the GPT-4o evaluation model share a provider family. To empirically eliminate potential bias rather than merely acknowledge it, we re-scored all 301 GPT-4o responses using `claude-haiku-4-5` as an independent cross-provider judge, applying the identical pre-registered scoring schema (Section 4).

Inter-judge agreement. Overall agreement between the GPT-4o-mini and Claude Haiku judges was 89.0% ($\kappa = 0.716$, substantial agreement by Landis–Koch conventions). Per-hop Cohen's κ: hop=1: 0.788; hop=2: 0.739; hop=3: 0.543; hop=4: 0.582. The lower κ at hops 3–4 reflects genuine criterion ambiguity for complex multi-hop responses rather than systematic provider bias.

Accuracy under each judge.

Judge	Hop=1	Hop=2	Hop=3	Hop=4
GPT-4o-mini	37.8%	26.1%	23.8%	14.7%
Claude Haiku	36.9%	32.6%	14.3%	15.7%

Both judges show the same monotone P1 pattern. The CA trend under the Claude Haiku judge ($z = -3.88$, $p < 0.0001$) is essentially identical to the original result ($z = -3.80$, $p < 0.0001$), confirming that the GPT-4o replication finding is independent of provider-family judge choice. No directional provider bias is detectable: the two judges disagree on 33/301 cases (11.0%), distributed roughly evenly across hops, with neither judge systematically favouring or penalising GPT-4o responses.

F.4 Dense Semantic RAG Baseline

To complement the BM25 lexical RAG baseline (Appendix F), we evaluated dense semantic retrieval using `all-MiniLM-L6-v2` sentence embeddings. For each question, we encoded all EHR chunks (200-character sliding windows, 50-character overlap) and retrieved the top-5 by cosine similarity to the question embedding, prepending them to the standard 8,000-character truncated EHR context. All other experimental parameters (model, judge, prompt) were identical to the main experiment.

Results. Accuracy by hop: hop=1 40.5% [31.9, 49.8], hop=2 26.1% [15.6, 40.3], hop=3 26.2% [15.3, 41.1], hop=4 20.6% [13.9, 29.4] (overall 29.6% [24.7, 35.0]; $n = 301$). Cochran–Armitage trend: $z = -3.13$, one-tailed $p < 0.001$. GEE hop$\times$condition interaction (dense RAG vs. zero-shot): OR $= 0.93$, 95% CI [0.73, 1.19], $p = 0.565$. The main hop effect persisted within the dense RAG condition (logistic OR per hop $= 0.73$ [0.60, 0.89], $p = 0.002$), confirming that semantic retrieval does not eliminate the compositional reasoning bottleneck.